Airline Customer Satisfaction Analysis Report

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1. Executive Summary

Date: March 2026
Data Analyzed: 25,976 passenger records
Overall Satisfaction Rate: 43.9%

This report identifies why 56% of passengers are dissatisfied and provides clear recommendations to improve customer experience.

2. Key Findings

Metric	Value	Status
Satisfied Customers	43.9%	Needs Improvement
Dissatisfied Customers	56.1%	High Concern
Loyal Customer Satisfaction	48.1%	Acceptable
Disloyal Customer Satisfaction	25.2%	Critical
Business Class Satisfaction	69.5%	Good
Economy Class Satisfaction	19.4%	Critical
Economy Plus Satisfaction	24.8%	Critical

3. The Three Critical Problems

PROBLEM #1: Inflight Wi-Fi - URGENT

Current WiFi Rating: 2.7/5 (Lowest of all services)

Why This Matters:

- Passengers who rate WiFi as 4 or 5 are **60-99% satisfied**
- Passengers who rate WiFi as 1-3 are only **24-34% satisfied**
- **54% of ALL dissatisfied customers** (7,891 people) rated WiFi poorly

```sql
--7: What's the satisfaction rate based on WiFi rating?
SELECT
    inflight_wifi_service as wifi_rating,
    COUNT(*) as passengers,
    ROUND(AVG(satisfaction) * 100, 2) as satisfaction_rate
FROM airline_satisfaction
GROUP BY inflight_wifi_service
ORDER BY inflight_wifi_service;
```

Data Output Messages Notifications

	wifi_rating bigint	passengers bigint	satisfaction_rate numeric
1	0	813	99.75
2	1	4488	33.91
3	2	6490	24.14
4	3	6317	25.69
5	4	4981	60.79
6	5	2887	98.79

Conclusion: Fixing WiFi alone could convert over 7,000 dissatisfied customers.

PROBLEM #2: Economy Class Experience

Economy Class Satisfaction: 19.4%

Who is affected:

- 11,564 Economy passengers (45% of all passengers)
- 1,917 Economy Plus passengers

Total affected: 13,481 passengers with satisfaction below 25%

```sql
--4: How does satisfaction vary by travel class?
select
    case
        when class = 0 then 'Business'
        when class = 1 then 'Economy'
        else 'Economy Plus'
    end as travel_class,
    count(*) as passengers,
    round(avg(satisfaction) * 100, 2) as satisfaction_rate
from airline_satisfaction
group by class
order by satisfaction_rate desc;
```

Data Output Messages Notifications

	travel_class text	passengers bigint	satisfaction_rate numeric
1	Business	12495	69.52
2	Economy Plus	1917	24.78
3	Economy	11564	19.39

The Breakdown:

Segment	Passengers	Current Satisfaction
Loyal Economy Customers	8,840	20.4%
Disloyal Economy Customers	2,724	16.1%
Economy Plus	1,917	24.8%

What Economy Passengers Hate Most:

- Seat comfort (gap of 0.92 between satisfied and dissatisfied)
- Food quality (gap of 0.58)
- Cleanliness rated 2.9/5 by dissatisfied customers
- Entertainment rated 2.9/5 by dissatisfied customers

```sql
--8: Which service has the biggest gap between satisfied and dissatisfied?
SELECT
    'Seat Comfort' as service,
    ROUND(AVG(CASE WHEN satisfaction = 1 THEN seat_comfort END), 2) as satisfied_avg,
    ROUND(AVG(CASE WHEN satisfaction = 0 THEN seat_comfort END), 2) as dissatisfied_avg,
    ROUND(AVG(CASE WHEN satisfaction = 1 THEN seat_comfort END) -
          AVG(CASE WHEN satisfaction = 0 THEN seat_comfort END), 2) as gap
FROM airline_satisfaction
UNION ALL
SELECT 'Food & Drink',
    ROUND(AVG(CASE WHEN satisfaction = 1 THEN food_and_drink END), 2),
    ROUND(AVG(CASE WHEN satisfaction = 0 THEN food_and_drink END), 2),
    ROUND(AVG(CASE WHEN satisfaction = 1 THEN food_and_drink END) -
          AVG(CASE WHEN satisfaction = 0 THEN food_and_drink END), 2)
FROM airline_satisfaction
ORDER BY gap DESC;
```

Data Output | Messages | Notifications

	service text	satisfied_avg numeric	dissatisfied_avg numeric	gap numeric
1	Seat Comfort	3.97	3.04	0.92
2	Food & Drink	3.54	2.96	0.58

PROBLEM #3: Flight Delays

Impact of Delays on Satisfaction:

Delay Duration	Satisfaction Rate
No Delay	46.7%
1-30 minutes	42.0%
31-60 minutes	36.6%
60+ minutes	36.4%

```sql
--9: How do delays affect satisfaction?
SELECT
    CASE
        WHEN departure_delay_in_minutes = 0 THEN 'No Delay'
        WHEN departure_delay_in_minutes BETWEEN 1 AND 30 THEN 'Minor (1-30 min)'
        WHEN departure_delay_in_minutes BETWEEN 31 AND 60 THEN 'Moderate (31-60 min)'
        ELSE 'Major (60+ min)'
    END as delay_category,
    COUNT(*) as flights,
    ROUND(CAST(AVG(satisfaction) * 100 AS numeric), 2) as satisfaction_rate
FROM airline_satisfaction
GROUP BY
    CASE
        WHEN departure_delay_in_minutes = 0 THEN 'No Delay'
        WHEN departure_delay_in_minutes BETWEEN 1 AND 30 THEN 'Minor (1-30 min)'
        WHEN departure_delay_in_minutes BETWEEN 31 AND 60 THEN 'Moderate (31-60 min)'
        ELSE 'Major (60+ min)'
    END
ORDER BY
    MIN(departure_delay_in_minutes);
```

Data Output | Messages | Notifications

	delay_category text	flights bigint	satisfaction_rate numeric
1	No Delay	14688	46.68
2	Minor (1-30 min)	7719	42.03
3	Moderate (31-60 min)	1833	36.61
4	Major (60+ min)	1736	36.35

The Tipping Point: Satisfaction drops below 40% after just **15 minutes** of delay.

Economy + Delay = Worst Case:

- Economy with no delay: 22.4% satisfied
- Economy with any delay: 15.5% satisfied

```sql
--13: What's the satisfaction rate for Economy class with delays?
SELECT
    CASE
        WHEN departure_delay_in_minutes = 0 THEN 'No Delay'
        ELSE 'With Delay'
    END as delay_status,
    COUNT(*) as passengers,
    ROUND(AVG(satisfaction) * 100, 2) as satisfaction_rate
FROM airline_satisfaction
WHERE class = 1  -- Economy class
GROUP BY delay_status;
```

Data Output | Messages | Notifications

	delay_status text	passengers bigint	satisfaction_rate numeric
1	No Delay	6592	22.35
2	With Delay	4972	15.47

Impact: 4,972 Economy passengers experience delays and only 15.5% are satisfied.

4. Secondary Issues

Service Ratings (1-5 scale)

Service	Rating	Status
Baggage Handling	3.63	Good
Seat Comfort	3.45	Needs Work
Entertainment	3.36	Needs Work
Cleanliness	3.29	Needs Work
Food & Drink	3.22	Poor
WiFi	**2.72**	**Critical**

```sql
--6: What are the average ratings for each service?
SELECT
    'Inflight WiFi' as service,
    ROUND(AVG(inflight_wifi_service), 2) as avg_rating
FROM airline_satisfaction
UNION ALL
SELECT 'Food & Drink', ROUND(AVG(food_and_drink), 2)
FROM airline_satisfaction
UNION ALL
SELECT 'Seat Comfort', ROUND(AVG(seat_comfort), 2)
FROM airline_satisfaction
UNION ALL
SELECT 'Inflight Entertainment', ROUND(AVG(inflight_entertainment), 2)
FROM airline_satisfaction
UNION ALL
SELECT 'Cleanliness', ROUND(AVG(cleanliness), 2)
FROM airline_satisfaction
UNION ALL
SELECT 'Baggage Handling', ROUND(AVG(baggage_handling), 2)
FROM airline_satisfaction
ORDER BY avg_rating DESC;
```

Data Output | Messages | Notifications

	service text	avg_rating numeric
1	Baggage Handling	3.63
2	Seat Comfort	3.45
3	Inflight Entertainment	3.36
4	Cleanliness	3.29
5	Food & Drink	3.22
6	Inflight WiFi	2.72

What Dissatisfied Customers Rate Lowest:

Service	Their Rating
WiFi	2.4/5
Entertainment	2.9/5
Cleanliness	2.9/5
Food	3.0/5
Seat Comfort	3.0/5

```sql
--12: Which services have the lowest ratings from dissatisfied customers?
SELECT
    ROUND(AVG(CASE WHEN satisfaction = 0 THEN inflight_wifi_service END), 2) as wifi_for_dissatisfied,
    ROUND(AVG(CASE WHEN satisfaction = 0 THEN seat_comfort END), 2) as seat_for_dissatisfied,
    ROUND(AVG(CASE WHEN satisfaction = 0 THEN food_and_drink END), 2) as food_for_dissatisfied,
    ROUND(AVG(CASE WHEN satisfaction = 0 THEN cleanliness END), 2) as clean_for_dissatisfied,
    ROUND(AVG(CASE WHEN satisfaction = 0 THEN inflight_entertainment END), 2) as entertainment_for_dissatisfied
FROM airline_satisfaction;
```

Data Output Messages Notifications

	wifi_for_dissatisfied numeric	seat_for_dissatisfied numeric	food_for_dissatisfied numeric	clean_for_dissatisfied numeric	entertainment_for_dissatisfied numeric
1	2.39	3.04	2.96	2.92	2.89

5. Customer Segments That Need Attention

Age Groups

Age Group	Passengers	Satisfaction	Priority
Under 18	1,916	18.1%	High
60+	2,496	26.8%	High
18-29	5,716	36.0%	Medium
30-39	5,045	41.7%	Low

```sql
--5: Which age groups are most/least satisfied?
select
    case
        when age < 18 then 'under 18'
        when age between 18 and 29 then '18-29'
        when age between 30 and 39 then '30-39'
        when age between 40 and 49 then '40-49'
        when age between 50 and 59 then '50-59'
        else '60+'
    end as age_group,
    count(*) as passengers,
    round(avg(satisfaction) * 100, 2) as satisfaction_rate
from airline_satisfaction
group by age_group
order by satisfaction_rate;
```

Data Output | Messages | Notifications

	age_group text	passengers bigint	satisfaction_rate numeric
1	under 18	1916	18.11
2	60+	2496	26.84
3	18-29	5716	35.97
4	30-39	5045	41.67
5	50-59	4863	57.58
6	40-49	5940	57.71

Total at-risk young and senior passengers: 10,128 people

Most Valuable Improvement Targets

Target Segment	Passengers	Current	Potential Gain
Loyal Economy	8,840	20.4%	+79.6%
Disloyal Economy	2,724	16.1%	+83.9%
Young Adults	5,716	36.0%	+64.0%
Seniors	2,496	26.8%	+73.2%

```sql
--17: Which passenger segment should we target first?
SELECT
    CASE
        WHEN customer_type = 0 THEN 'Loyal' ELSE 'Disloyal'
    END as loyalty,
    CASE
        WHEN class = 0 THEN 'Business'
        WHEN class = 1 THEN 'Economy'
        ELSE 'Eco Plus'
    END as class,
    COUNT(*) as passengers,
    ROUND(AVG(satisfaction) * 100, 2) as current_satisfaction,
    ROUND(100 - AVG(satisfaction) * 100, 2) as improvement_potential
FROM airline_satisfaction
GROUP BY loyalty, class
ORDER BY improvement_potential DESC
LIMIT 5;
```

Data Output | Messages | Notifications

	loyalty text	class text	passengers bigint	current_satisfaction numeric	improvement_potential numeric
1	Disloyal	Eco Plus	200	8.50	91.50
2	Disloyal	Economy	2724	16.08	83.92
3	Loyal	Economy	8840	20.41	79.59
4	Loyal	Eco Plus	1717	26.67	73.33
5	Disloyal	Business	1875	40.16	59.84

6. What Factors Matter Most?

Correlation with Satisfaction (Higher = More Important):

Factor	Importance
Entertainment	0.40
Seat Comfort	0.35
Cleanliness	0.31
WiFi	0.28
Food & Drink	0.22

```sql
--14: Top 3 factors that would most improve satisfaction (correlation)
SELECT
    'WiFi' as factor,
    ROUND(CORR(inflight_wifi_service, satisfaction)::numeric, 3) as correlation_with_satisfaction
FROM airline_satisfaction
UNION ALL
SELECT 'Seat Comfort', ROUND(CORR(seat_comfort, satisfaction)::numeric, 3)
FROM airline_satisfaction
UNION ALL
SELECT 'Food & Drink', ROUND(CORR(food_and_drink, satisfaction)::numeric, 3)
FROM airline_satisfaction
UNION ALL
SELECT 'Entertainment', ROUND(CORR(inflight_entertainment, satisfaction)::numeric, 3)
FROM airline_satisfaction
UNION ALL
SELECT 'Cleanliness', ROUND(CORR(cleanliness, satisfaction)::numeric, 3)
FROM airline_satisfaction
ORDER BY correlation_with_satisfaction DESC;
```

Data Output | Messages | Notifications ...

	factor text	correlation_with_satisfaction numeric
1	Entertainment	0.399
2	Seat Comfort	0.346
3	Cleanliness	0.314
4	WiFi	0.280
5	Food & Drink	0.217

7. Recommendations

Immediate Actions (Next 3 Months)

Priority	Action	Target	Expected Impact
1	**Upgrade Wi-Fi**	All flights	+15-20% satisfaction
2	**Improve Economy seats**	11,500 passengers	+10-15% satisfaction
3	**Better food in Economy**	Economy class	+5-8% satisfaction
4	**15-minute delay guarantee**	All flights	+5% satisfaction

Wi-Fi Upgrade Details:

- Free basic WiFi for all passengers
- 3x faster speeds
- Target: Increase WiFi rating from 2.7 to 4.0+

Economy Seat Improvements:

- Add 2 inches leg room where possible
- Replace worn seats
- Ensure all seats recline

Short-Term Actions (3-6 Months)

Action	Target	Expected Impact
Young passenger program (modern entertainment, charging ports)	5,700 young adults	+8-10%
Senior assistance program (priority boarding, extra help)	2,500 seniors	+8-10%
Entertainment system update	All flights	+5-8%
Deep cleaning initiative	All flights	+5%

Long-Term Actions (6-12 Months)

Action	Target	Expected Impact
Enhanced loyalty rewards for Economy	8,800 loyal Economy	+10-12%
Delay compensation policy	All delayed flights	+8-10%
Economy Plus experience upgrade	1,900 passengers	+10-15%

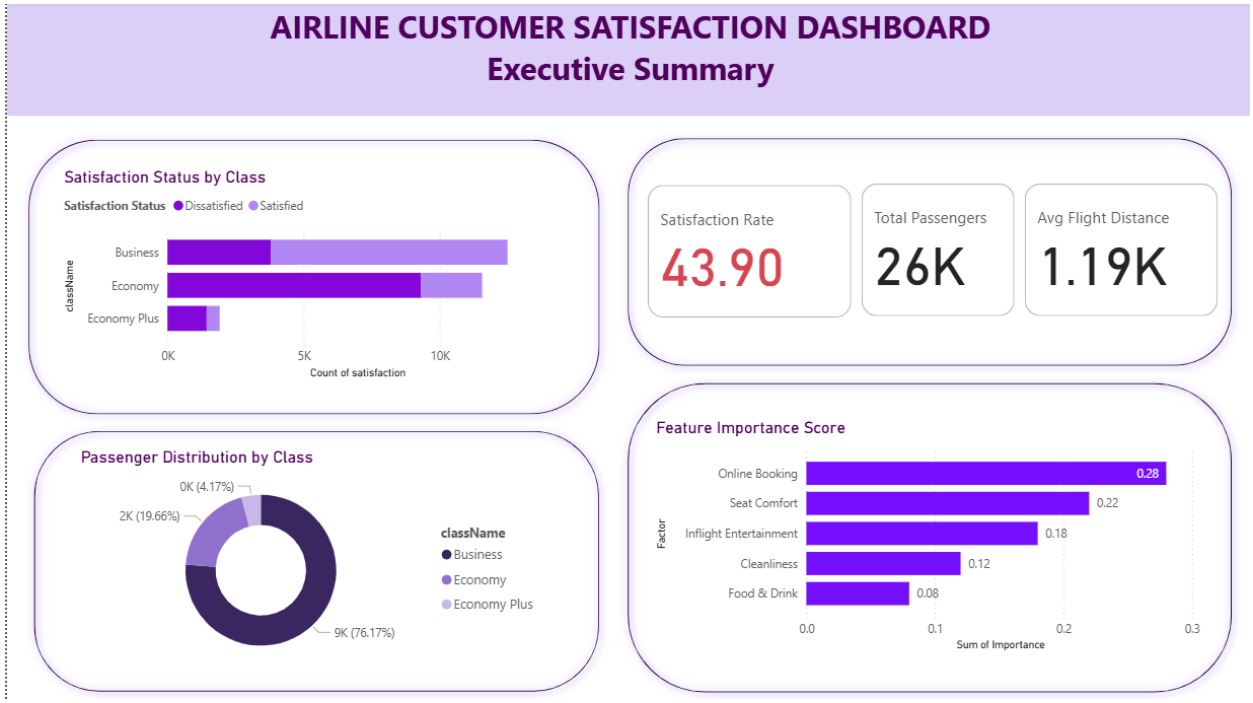
8. Projected Improvement

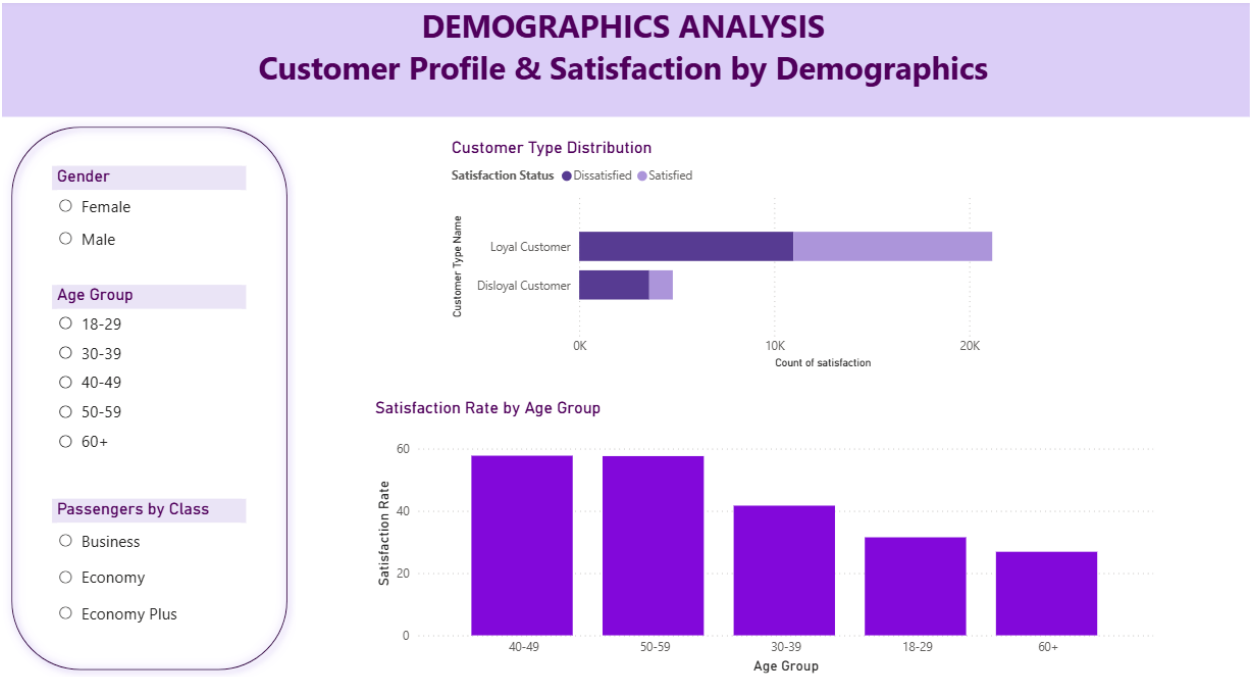
Area	Current	Target	Gain
Overall Satisfaction	43.9%	**60%**	**+16.1%**
Satisfied Passengers	11,403	15,586	+4,183
WiFi Rating	2.7/5	4.0/5	+1.3
Economy Satisfaction	19.4%	35%	+15.6%

9. Quick Reference Card

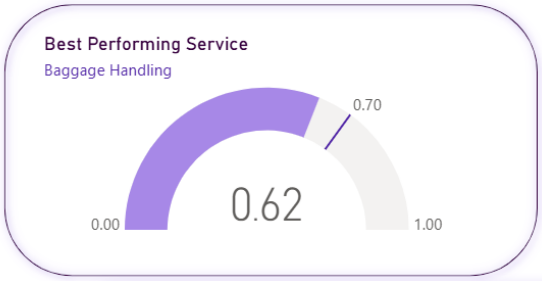
Problem	Impact	Solution	Timeline
Poor WiFi	7,891 dissatisfied	Free fast Wi-Fi	0-3 months
Economy Class	13,481 @ <25% sat	Better seats & food	3-6 months
Delays	15 min = 40% sat	15-min guarantee	0-3 months
Young passengers	5,716 @ 36% sat	Modern amenities	3-6 months
Seniors	2,496 @ 27% sat	Assistance program	3-6 months

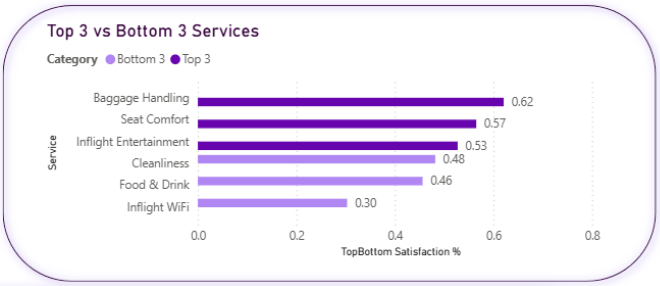
10. Appendix

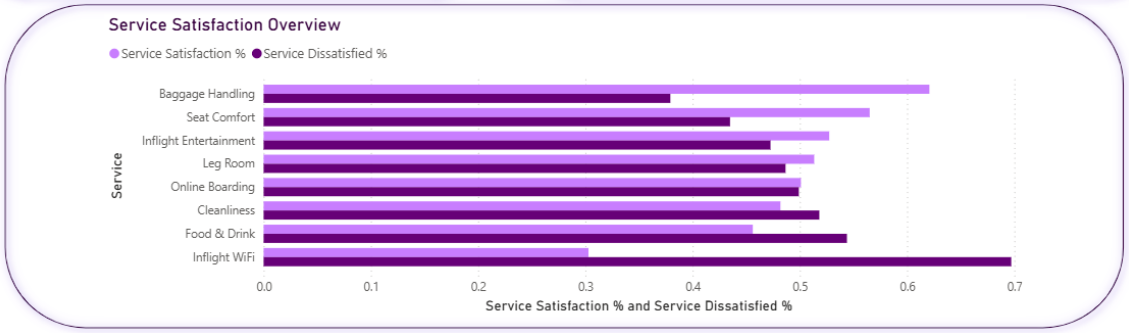
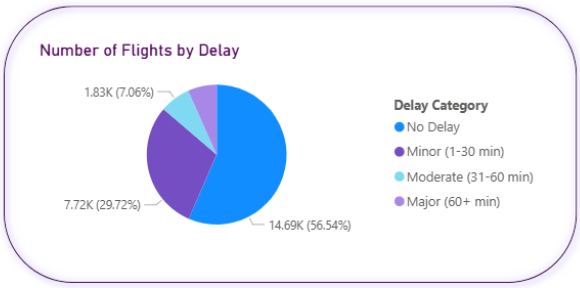
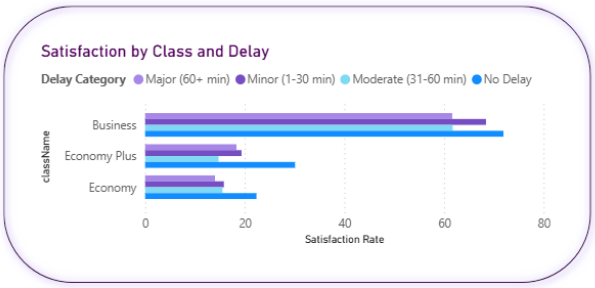
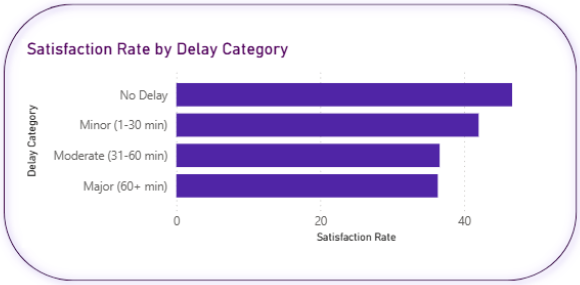
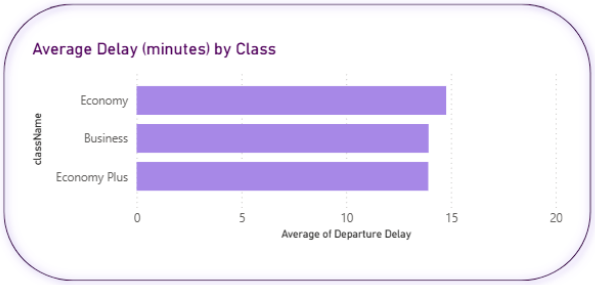

AIRLINE CUSTOMER SATISFACTION DASHBOARD
Executive Summary
Satisfaction Status by Class
Satisfaction Status Dissatisfied Satisfied
className
Business
Economy
Economy Plus
0K
5K
10K
Count of satisfaction
Satisfaction Rate
43.90
Total Passengers
26K
Avg Flight Distance
1.19K
Passenger Distribution by Class
0K (4.17%)
2K (19.66%)
9K (76.17%)
className
Business
Economy
Economy Plus
Feature Importance Score
Online Booking
0.28
Seat Comfort
0.22
Inflight Entertainment
0.18
Cleanliness
0.12
Food & Drink
0.08
Factor
0.0
0.1
0.2
0.3
Sum of Importance


DEMOGRAPHICS ANALYSIS
Customer Profile & Satisfaction by Demographics
Gender
Female
Male
Age Group
18-29
30-39
40-49
50-59
60+
Passengers by Class
Business
Economy
Economy Plus
Customer Type Distribution
Satisfaction Status Dissatisfied Satisfied
Customer Type Name
Loyal Customer
Disloyal Customer
0K
10K
20K
Count of satisfaction
Satisfaction Rate by Age Group
Satisfaction Rate
60
40
20
0
40-49
50-59
30-39
18-29
60+
Age Group

SERVICE QUALITY DASHBOARD







DELAY IMPACT ANALYSIS